

SARATHKUMAR RAVICHANDRAN

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Coimbatore, Tamil Nadu

[LinkedIn](#)[GitHub](#)[LeetCode](#) (550+ Solved)

EDUCATION

B.Tech Information Technology, College of Engineering Guindy, Anna University, Chennai 2023 – 2027
CGPA: 8.82

HSC, Suguna RIP V Hr Sec School, Coimbatore 2021 – 2023
Percentage: 98.5% | JEE Mains: 94.88 %ile | Student Council President

SKILLS

Programming: C, C++, Python, Java (Fundamentals)
AI/ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), Image Processing
Web Development: HTML, CSS, JavaScript, EJS, Node.js, Express.js, PostgreSQL
Tools: VS Code, Git, GitHub, Postman
Soft Skills: Extempore, Public Speaking, Teaching, Leadership, Event Management, Tamil Poem Writing

PROJECTS

IoT Smart Waste Segregator with Predictive Monitoring (*ESP32, IoT, ML*) [Demo Video](#) | [GitHub](#)

- Developed an ESP32-based smart waste monitoring system for smart-city municipality use, integrating ultrasonic sensing, moisture-based wet/dry waste segregation, and real-time WiFi alerts, while building ML models to predict waste accumulation and optimize pickup scheduling, achieving **R² up to 0.95 with RMSE 0.14–0.17**.

AI-Powered Medical Prescription Analyzer (*Deep Learning, OCR, LLMs*) [Website](#) | [GitHub](#)

- Built an end-to-end prescription intelligence platform that converts illegible handwritten medical prescriptions into structured digital text using custom-trained OCR, improving recognition accuracy through domain-specific training on 1,000+ medical terms and leveraging LLMs for prescription interpretation and patient assistance.

C³ – CEG Club Connex (*Node.js, Express.js, PostgreSQL*) [GitHub](#)

- Developed a centralized platform for managing CEG clubs, enabling students to track academic and extracurricular participation in one system, and implemented automated PDF generation to produce personalized performance tracker reports for streamlined progress analysis and record management.

C-CargoDS Manager (*C, Graphviz API, DSA*) [GitHub](#)

- Developed a cargo management system in C using stacks, binary search trees, and Dijkstra's algorithm, with Graphviz-based route visualization for efficient route planning and logistics management.

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

Top 5 Finalist – Societal Oriented Project 2025 (Musicio)

Ranked among the top 5 teams out of 50 for designing an intelligent music-learning platform leveraging machine learning to analyze instrument notes and deliver real-time feedback and personalized learning experiences.

Winner, Win With Words – Toastmasters CEG, Techofes 2024

Secured 1st place in a multi-round public speaking competition featuring extempore, debate, and life rendition speech.

Secretary, AU Podium – Official Speech Club, Anna University

Joint Secretary, NSS Unit - 4, Anna University

Master of Ceremonies, College Events

Hosted large-scale events including *Infiesta 2025*, *Freshers' Day*, *Aatral*, and *Anthaathi*, managing audience engagement and event flow for 500+ attendees.

CERTIFICATIONS

Top 1% Elite Scorer, NPTEL – Python for Data Science - IIT Madras ([link](#))

Full-Stack Web Development Bootcamp – Udemy ([link](#))

Completed a 62-hour project-based full-stack development bootcamp under Dr. Angela Yu.

AI Prompt Engineering Bootcamp – Udemy ([link](#))

Gained practical exposure to Prompt Engineering, AI Agents, RAG pipelines, and LLM-powered applications, with an understanding of the end-to-end workflow behind modern generative AI systems.